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THE MOLECULAR MODEL

1.1 Introduction

A gas flow may be modelled at either the macroscopic or the microscopic level. The macroscopic model regards the gas as a continuous medium and the description is in terms of the spatial and temporal variations of the familiar flow properties such as the velocity, density, pressure, and temperature. The Navier–Stokes equations provide the conventional mathematical model of a gas as a continuum. The macroscopic properties are the dependant variables in these equations, while the independent variables are the spatial coordinates and time.

The microscopic or molecular model recognizes the particulate structure of the gas as a myriad of discrete molecules and ideally provides information on the position, velocity, and state of every molecule at all times. The mathematical model at this level is the Boltzmann equation. This has the fraction of molecules in a given location and state as its single dependent variable, but the independent variables are increased by the number of physical variables on which the state depends. In the simplest case of a monatomic gas with no internal degrees of freedom, the additional dimensions of this *phase space* are the three velocity components of the molecules. A one-dimensional steady flow of such a gas becomes a three-dimensional problem in phase space (the velocity distribution is axially symmetric about the velocity component in the flow direction), while a two-dimensional steady flow becomes five-dimensional. This means that the Boltzmann equation is not amenable to analytical solution for non-trivial problems, and it presents overwhelming difficulties to conventional numerical methods. However, the discrete structure of the gas at the molecular level enables these difficulties to be circumvented through direct physical, rather than mathematical, modelling.

This book is concerned with the analysis of gas flows at the microscopic or molecular level, with particular emphasis on computational methods that employ the physical or direct simulation approach. While this physical approach has some advantages over the traditional mathematical formulation for computational studies, notably the absence of numerical instabilities, the molecular model is generally more demanding of computer resources. It is therefore important to delineate the circumstances under which the continuum model loses its validity and must be replaced by the molecular model.

1.2 The requirement for a molecular description

The macroscopic properties may be identified with average values of the appropriate molecular quantities at any location in a flow. They may therefore be defined as long as there are a sufficient number of molecules within the smallest significant volume of a flow. This condition is almost always satisfied and the results from the molecular model can therefore be expressed in terms of the familiar continuum flow properties. Moreover, the equations that express the conservation of mass, momentum, and energy in a flow are common to, and can be derived from, either model. While this might suggest that neither of the approaches can provide information that is not also accessible to the other, it must be remembered that the conservation equations do not form a determinate set unless the shear stresses and heat flux can be expressed in terms of the lower-order macroscopic quantities. It is the failure to meet this condition, rather than the breakdown of the continuum description, which imposes a limit on the range of validity of the continuum equations. More specifically, the transport terms in the Navier–Stokes equations of continuum gas dynamics fail when gradients of the macroscopic variables become so steep that their scale length is of the same order as the average distance travelled by the molecules between collisions, or *mean free path*.

The degree of rarefaction of a gas is generally expressed through the *Knudsen number* (*Kn*) which is the ratio of the mean free path λ to the characteristic dimension L, i.e.

$$(Kn) = \lambda/L. \tag{1.1}$$

The traditional requirement for the Navier–Stokes equations to be valid is that the Knudsen number should be less than 0.1. This can be misleading if L is chosen to be some overall dimension of the flow in order to define a single *overall* Knudsen number for the complete flow. The limit can be specified precisely if a *local* Knudsen number is defined with L as the scale length of the macroscopic gradients; e.g.

$$L = \frac{\rho}{\mathrm{d}\rho/\mathrm{d}x}. \tag{1.2}$$

The error in the Navier–Stokes result is significant in the regions of the flow where the appropriately defined local number exceeds 0.1, and the upper limit on the local Knudsen number at which the continuum model must be replaced by the molecular model may be taken to be 0.2.

The transport terms vanish in the limit of zero Knudsen number and the Navier–Stokes equations then reduce to the inviscid Euler equations. The flow is then isentropic from the continuum viewpoint, while the equivalent molecular viewpoint is that the velocity distribution function is everywhere of the local equilibrium or Maxwellian form. The opposite limit of infinite Knudsen number is the *collisionless* or *free-molecule* flow regime. These Knudsen number limits on the conventional mathematical formulations are shown schematically in Fig. 1.1.

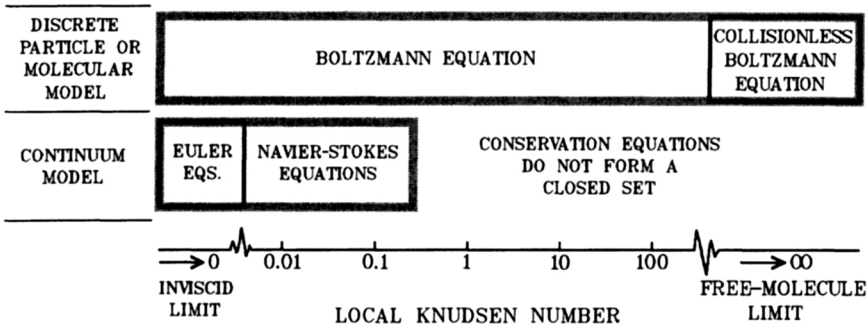


FIG. 1.1 The Knudsen number limits on the mathematical models.

A major achievement of classical kinetic theory, as described by Chapman and Cowling (1952), was the development of the Chapman–Enskog theory for the coefficients of viscosity, heat conduction, and diffusion. This theory authenticated the assumption, inherent in the Navier–Stokes formulation, that the shear stresses, heat fluxes, and diffusion velocities are linear functions of the gradients in velocity, temperature, and species concentration. The Chapman–Enskog theory also established the limits of validity of the formulation because it assumes that the velocity distribution function f is a small perturbation of the equilibrium or Maxwellian function f_0. For example, it will be shown in Chapter 4 that, for the special case of a flow in the x-direction with gradients only in the y-direction, the Chapman–Enskog distribution function can be written

$$f = f_0\left[1 - C\beta v'\left\{3\left(\beta^2 c'^2 - \frac{5}{2}\right)\frac{\lambda}{T}\frac{\partial T}{\partial y} + 4\beta u' s\,\frac{\lambda}{u_0}\frac{\partial u_0}{\partial y}\right\}\right], \tag{1.3}$$

where C is a numerical factor that depends on the gas. Note that the local Knudsen numbers based on the stream velocity and temperature appear explicitly in the terms that are attributable to the shear stress and the heat flux, respectively. Since the Chapman–Enskog result is the first term in a series expansion about these local Knudsen numbers, the theory is valid only when they are small in comparison with unity.

The transport properties are most significant within boundary layers and shock waves. The mean free path is inversely proportional to the gas density and, for a given shock strength, the width of the wave is also inversely proportional to the density. The local Knudsen numbers within the wave are therefore independent of the density and the validity of the Navier–Stokes equations depends only on the shock Mach number. It has been found that the restriction on the validity of the Navier–Stokes solutions for shock wave structure is that the shock Mach number should be significantly below two. In the case of a laminar boundary layer in a low-speed flow, the thickness of the layer is inversely proportional to the

square root of the density. The gradients therefore decrease less rapidly than the mean free path increases as the density falls, and the local Knudsen numbers become larger at low densities. The increase in the thickness of both these flow features means that, in a flow with shock waves and/or boundary layers, the proportion of the flow that is viscous increases as the flow becomes rarefied. At extremely low densities, the shock waves and boundary layers first merge and then lose their identity as free-molecule conditions are approached.

Outside boundary layers and shock waves, continuum flow is assumed to be isentropic and it might be thought that the Euler equations yield correct results at all Knudsen numbers. The gradients of the macroscopic properties in an isentropic flow depend only on the size of the flow. However, these changes in the macroscopic flow properties can occur only through intermolecular collisions and, as the density falls, the collision rate becomes too low for the maintenance of these gradients. The continuum model then breaks down, with the initial symptom being an anisotropic pressure tensor. The occurrence of this type of breakdown in gaseous expansions was studied by Bird (1970) and it was found to correlate with the 'breakdown parameter'

$$P = \frac{1}{v}\left|\frac{D(\ln\rho)}{Dt}\right|. \tag{1.4}$$

For a steady flow, this equation can be written

$$P = \frac{\pi^{1/2}}{2} s \frac{\lambda}{\rho}\left|\frac{d\rho}{dx}\right|, \tag{1.5}$$

and we recover a local Knudsen number similar to those in eqn (1.3). The initial breakdown in both steady and unsteady expansions has been found to correlate with a value of P of approximately 0.02.

A large Knudsen number may result from either a large mean free path or a small scale length of the macroscopic gradients. The former is usually the case and is a consequence of a very low gas density. This is why the title 'rarefied gas dynamics' is frequently applied to the subject matter of this book. However, it must be kept in mind that the alternative requirement of a small characteristic dimension can be met at any density. It has already been noted that the internal structure of a strong shock wave appears to require the molecular model at all densities. Similarly, the molecular approach is required for the study of the forces on a very small particle suspended or moving through the atmosphere, or for the propagation of sound at extremely high frequencies.

1.3 The simple dilute gas

The basic quantities associated with the molecular model are the number of molecules per unit volume and the mass, size, velocity, and internal state of each molecule. These quantities must be related to the mean free path and collision frequency in order to establish the distance and time

scales of the effects due to the collisional interactions among the molecules. Also, since the results from the molecular approach will generally be presented in terms of the macroscopic quantities, we must establish the formal relationships between the microscopic and macroscopic quantities. For reasons of simplicity and clarity, the discussion in this section will be restricted to a gas consisting of a single chemical species in which all molecules are assumed to have the same structure. Such a gas is called a *simple gas*.

The number of molecules in one mole of a gas is a fundamental physical constant called *Avogadro's number* $\mathcal{N}$. Avogadro's law also states that the volume occupied by one mole of any gas at a particular temperature and pressure is the same for all gases. The number of molecules per unit volume, or *number density* n, of a gas depends on the temperature and pressure, but is independent of the composition of the gas. The mass m of a single molecule is obtained by dividing the *molecular weight* $\mathcal{M}$ of the gas by Avogadro's number, i.e.

$$m = \mathcal{M}/\mathcal{N} = \mathcal{M}\, m_u, \tag{1.6}$$

where m_u is the atomic mass constant. The average volume available to a molecule is $1/n$, so the mean molecular spacing δ is given by

$$\delta = n^{-1/3} . \tag{1.7}$$

A hard elastic sphere of diameter d provides an over-simplified but useful model of a molecule. Then, as shown in Fig. 1.2, two molecules collide if their trajectories are such that the distance between their centres decreases to d. The *total collision cross-section* for these molecules is therefore

$$\sigma_T = \pi d^2 . \tag{1.8}$$

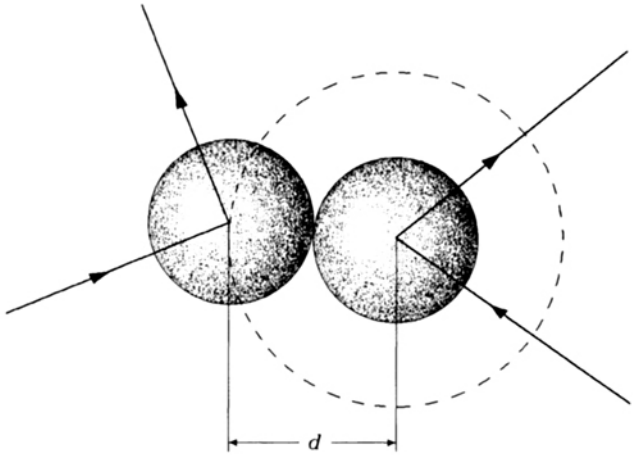


Fig. 1.2 Collision between two hard spheres of diameter d.

The word 'molecule' is used as a generic term and includes both monatomic molecules consisting of a single atom, diatomic molecules containing two atoms, and polyatomic molecules that contain more than two atoms. Each atom of a real molecule consists of a nucleus surrounded by orbiting electrons. Molecular size is a quantity that cannot be precisely and uniquely defined and it will be necessary to qualify the results from elementary kinetic theory that depend on the molecular 'diameter'.

Classical studies of the effects of intermolecular collisions are based on the force fields of the molecules. These fields are assumed to be spherically symmetric and the general form of the force between two neutral molecules is shown in Fig. 1.3 as a function of the distance between the nuclei. The force is effectively zero at large distances; it becomes weakly attractive when the molecules are sufficiently close for the interaction to commence, but then decreases again to become very strongly repulsive at short distances. The intermolecular forces in a diatomic or polyatomic gas are a function of the orientation of the molecules. However, in view of the random orientation in the extremely large number of collisions that occur in almost all cases, it is reasonable to assume spherically symmetric fields for these gases. Collision dynamics will be dealt with in detail in §2.2 where procedures for the calculation of collision cross-sections and molecular diameters will be discussed. The collision cross-section of realistic models is a function of the relative speed between the molecules and experience has shown that it is important to reproduce this behaviour. The form of the force field and the consequent distribution of scattering angles in the collisions is comparatively unimportant.

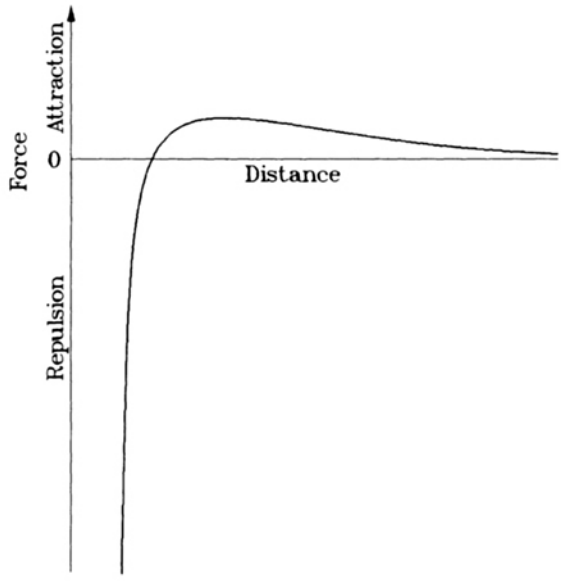


FIG. 1.3 Typical intermolecular force field.

The proportion of the space occupied by a gas that actually contains a molecule is of the order of $(d/\delta)^3$. Eqn (1.7) shows that, for sufficiently low densities, the molecular spacing δ is large compared with the effective molecular diameter d. Under these circumstances, only an extremely small proportion of space is occupied by molecules and each molecule will, for the most part, be moving outside the range of influence of other molecules. Moreover, when it does suffer a collision, it is overwhelmingly likely to be a *binary collision* involving only one other molecule. This situation may be characterized by the condition

$$\delta \gg d, \tag{1.9}$$

and defines a *dilute gas*. The time-scale of the macroscopic processes is set by the *mean collision time* which is, by definition, the mean time between the successive collisions suffered by any particular molecule. The reciprocal of this quantity is in more common use and is called the *mean collision rate* or *collision frequency* ν per molecule. In the derivation of an expression for this quantity we will fix our attention on a particular molecule which will be referred to as the *test* molecule. The velocities of the other, or *field*, molecules are distributed in some unspecified manner. Consider those field molecules with velocity between $\boldsymbol{c}$ and $\boldsymbol{c}+\Delta\boldsymbol{c}$. These will be referred to as molecules of class $\boldsymbol{c}$ and their number density is denoted by Δn. If the velocity of the test molecule is $\boldsymbol{c}_t$, the relative velocity between the test molecule and the field molecules of class $\boldsymbol{c}$ is $\boldsymbol{c}_r = \boldsymbol{c}_t - \boldsymbol{c}$. Now choose a frame of reference in which the test molecule moves with velocity $\boldsymbol{c}_r$ while the field molecules of class $\boldsymbol{c}$ are stationary. Then, over a time interval Δt much shorter than the mean collision time, the test molecule would collide with any field molecule which has its centre within the cylinder of volume $\sigma_T c_r \Delta t$, as shown in Fig. 1.4.

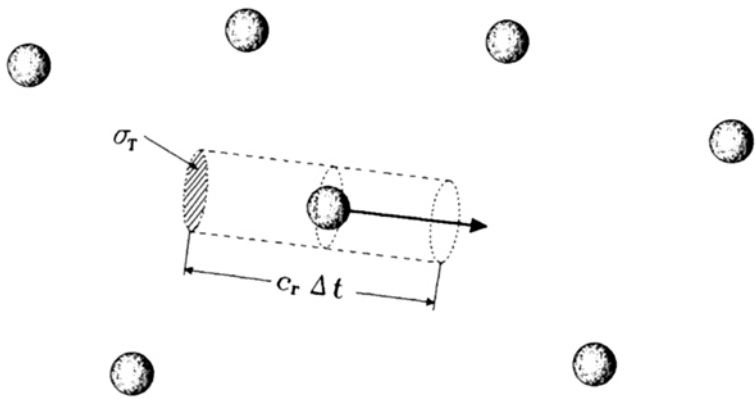


Fig. 1.4 Effective volume swept out by moving test molecule among stationary field molecules.

The probability of a collision between the test molecule and a molecule of class $\boldsymbol{c}$ in the time interval Δt is therefore $\Delta n\, \sigma_T c_r \Delta t$. When collisions do occur, the cylinder swept out by the collision cross-section along the trajectory becomes distorted. However, for a dilute gas in which only a very small proportion of the trajectory is affected by collisions, the restriction on Δt can be removed and the number of collisions per unit time with a class $\boldsymbol{c}$ molecule is $\Delta n\, \sigma_T c_r$. The mean collision rate is obtained by summing over all velocity classes and therefore over all values of c_r. That is,

$$\nu = \sum (\Delta n \sigma_T c_r) = n \sum [(\Delta n / n) \sigma_T c_r]$$

and, since $\Delta n/n$ is the fraction of molecules with cross-section σ_T and relative velocity $\boldsymbol{c}_r$,

$$\nu = n\, \overline{\sigma_T c_r}\,. \tag{1.10}$$

A bar over a quantity or expression denotes the average value over all molecules in the sample. For hard sphere molecules, this becomes

$$\nu = \sigma_T n\, \overline{c_r} = \pi d^2 n\, \overline{c_r}\,. \tag{1.10a}$$

The total number of collisions per unit time per unit volume of gas is therefore given by

$$N_c = \tfrac{1}{2} n \nu = \tfrac{1}{2} n^2 \overline{\sigma_T c_r}. \tag{1.11}$$

The symmetry factor of one half is introduced because each collision involves two molecules.

The *mean free path* is the average distance travelled by a molecule between collisions. It is defined in a frame of reference moving with the stream speed of the gas and is therefore equal to the mean thermal speed $\overline{c'}$ of the molecule divided by the collision frequency, i.e.

$$\lambda = c'/\nu = [\, n\, (\overline{\sigma_T c_r} / \overline{c'})\,]^{-1} \tag{1.12}$$

or, for the constant cross-section hard sphere case,

$$\lambda = [\, (\overline{c_r} / \overline{c'})\, \pi d^2 n\,]^{-1}\,. \tag{1.12a}$$

Before moving on to discuss the formal relationships between the microscopic and macroscopic properties, we must elaborate on the concept of *equilibrium* as applied to gases. Consider a volume of gas that is completely isolated from any outside influence. If this gas remains undisturbed for a time that is sufficiently long in comparison with the mean collision time, it may be regarded as being in an equilibrium state. Then, if the number of molecules in the volume is sufficiently large that statistical fluctuations may be neglected, there are no gradients in the macroscopic properties with either distance or time. Also, the fraction of molecules in any velocity class remains constant with time, even though the velocity of an individual

molecule changes with each of its collisions. It is obvious that there can be no preferred direction in an equilibrium gas and the velocity distribution of the molecules must be isotropic. The form of the velocity distribution in an equilibrium gas will be derived in Chapter 3 and its properties will be discussed in Chapter 4. If the macroscopic gradients in a gas flow are sufficiently small and the collision rate is sufficiently high, the velocity distribution of each element of the gas adjusts to the equilibrium state appropriate to the local macroscopic properties as it moves through the gas. At the microscopic level, the flow can then be regarded as being in *local thermodynamic equilibrium* and, at the macroscopic or continuum level, this is equivalent to *isentropic* flow. As noted earlier, the continuum approach is valid only for flows in which the departure from equilibrium is small, but the definitions of the macroscopic flow properties that are presented in the following sections remain valid for any degree of non-equilibrium.

1.4 Macroscopic properties in a simple gas

The first of the macroscopic properties to be discussed is the *density* ρ. This is defined as the mass per unit volume of the gas, and is therefore equal to the product of the number of molecules per unit volume and the mass of an individual molecule, i.e.

$$\rho = n\,m\,. \tag{1.13}$$

In the continuum model, macroscopic quantities such as the density are associated with a 'point' in a flow. Values at a 'point' must, in fact, be based on the molecules within a small volume element that encloses the point. An element of volume V contains a number N of molecules and this number is subject to statistical fluctuations about the mean value nV. The probability $P(N)$ of a particular value of N is given by the Poisson distribution

$$P(N) = (nV)^N \exp(-nV)/N!\,. \tag{1.14}$$

For large values of nV, this distribution becomes indistinguishable from a normal or Gaussian distribution

$$P(N) \approx (2\pi nV)^{-\frac{1}{2}} \exp[-(N-nV)^2/(2nV)]\,. \tag{1.15}$$

The integration of this distribution shows that the probability of an individual sample falling within $A\sqrt{nV}$ from the average nV is $\mathrm{erf}(A/\sqrt{2})$. The standard deviation of the fluctuations is therefore $1/\sqrt{nV}$. The expected magnitude of the fluctuations is shown in Fig. 1.5. Eqn (1.7) enables nV to be written V/δ^3, where δ is the mean molecular spacing. Therefore, for a meaningful result, the establishment of a macroscopic quantity by averaging over the molecules in a small element of volume requires that the typical dimension $V^{1/3}$ of the element should satisfy the condition.

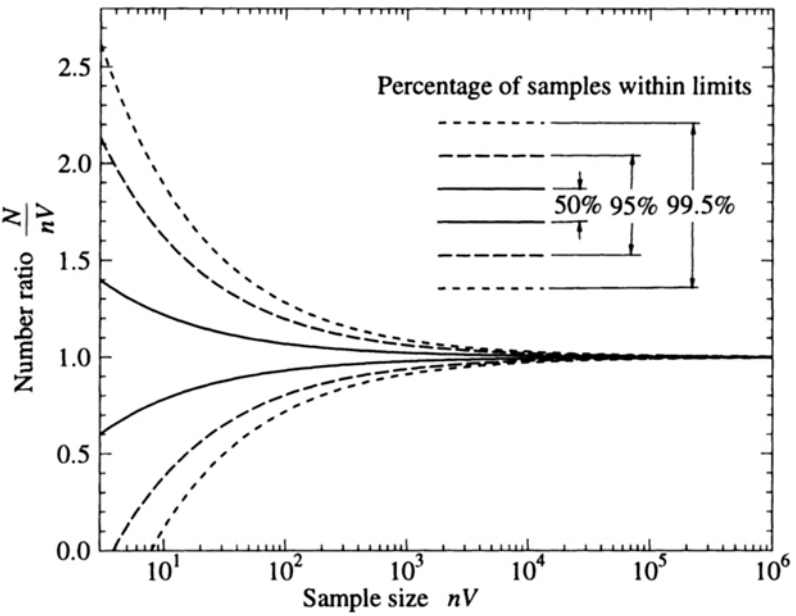


Fig. 1.5 Statistical fluctuations as a function of the sample size.

$$V^{1/3} >> \delta \, . \tag{1.16}$$

Unless the dimensions of the volume element are small compared with the scale length L of the macroscopic gradients, the macroscopic properties based on the molecules in the element will depend on the size of the element. For a three-dimensional flow with gradients in all directions, this leads to the requirement that $V^{1/3}$ should be much smaller than L. However, for one-dimensional and two-dimensional flows, the volume element may be indefinitely elongated in the direction or directions with zero gradients. It is always possible in such flows to define an element with a dimension in the direction of a gradient small compared with the scale length L and the volume sufficiently large to define a meaningful average, even though $V^{1/3}$ may be larger than L.

The above discussion deals only with *instantaneous averages* in a single flow or system. Two additional types of average must be taken into account before conclusions can be made about the conditions that allow meaningful macroscopic quantities to be defined for a flow. The first is established by summing the appropriate properties of the molecules in the volume element over an extended time interval. This is called the *time average* and enables any steady flow to be described in terms of the macroscopic properties. The second average is an instantaneous average taken over the molecules in corresponding volume elements in an

indefinitely large number of similar systems. This *ensemble average* can be established whenever an experiment or calculation can be repeated indefinitely. Time or ensemble averages enable macroscopic properties to be established for almost any flow, although the description is essentially probabilistic and fluctuations may have to be taken into account. When both time and ensemble averages can be used to describe a flow, the two descriptions can be expected to be identical. The molecular motion is then said to be *ergodic*.

The remaining macroscopic properties of interest are related to the transport of mass, momentum, and energy in the flow as a result of the molecular motion. Before proceeding with the discussion of these quantities, we must establish an important general result for the flux of some quantity Q across a small element of area at some location in the gas. This relates the flux to averages taken over the molecules in a volume element at the same location. The element of area has the magnitude ΔS and unit normal vector $\boldsymbol{e}$, as shown in Fig. 1.6. Consider the molecules of class $\boldsymbol{c}$ which have a velocity between $\boldsymbol{c}$ and $\boldsymbol{c}+\Delta\boldsymbol{c}$ and let their number density again be Δn. The molecules of this class that cross ΔS in a short time interval Δt are contained, at the beginning of Δt, within the cylinder projected from ΔS in the direction opposite to $\boldsymbol{c}$ with length $c\,\Delta t$. The height of this cylinder when measured in the direction of $\boldsymbol{e}$, normal to ΔS, is $\boldsymbol{c}.\boldsymbol{e}\,\Delta t$, and its volume is $\boldsymbol{c}.\boldsymbol{e}\,\Delta t\,\Delta S$. The quantity Q is associated with each molecule and is either a constant or a function of $\boldsymbol{c}$. The flux of Q across the element per unit area per unit time in the direction of $\boldsymbol{e}$ is, therefore, $\Delta n\, Q\,\boldsymbol{c}.\boldsymbol{e}$. The total flux is obtained by summing over all velocity classes and can be written

$$n\sum\left[(\Delta n/n)\,Q\,\boldsymbol{c}.\boldsymbol{e}\right]$$

or

$$n\,\overline{Q\,\boldsymbol{c}.\boldsymbol{e}}\,. \tag{1.17}$$

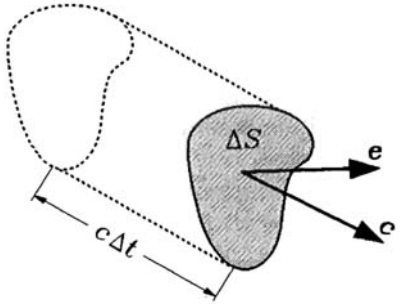


Fig. 1.6 Flux of molecules of class $\boldsymbol{c}$ across an element of area ΔS.

Note that the expression (1.17) is for the total flux and includes the contributions from molecules crossing the element of area in both the positive and negative $\boldsymbol{e}$ directions. We will sometimes require the flux in one direction only and, without loss of generality, this may be taken in the positive $\boldsymbol{e}$ direction. The required flux is then obtained by considering only those molecules with a positive value of $\boldsymbol{c}.\boldsymbol{e}$. The flux in the positive $\boldsymbol{e}$ direction is, therefore,

$$n(\overline{Q\boldsymbol{c}.\boldsymbol{e}})_{\boldsymbol{c}.\boldsymbol{e}>0}. \tag{1.18}$$

It has been assumed in the derivation of expressions (1.17) and (1.18) that the quantity Q is transferred across the element of area ΔS only when the molecule with which it is associated crosses the element. This is exact for the number flux and mass flux, but is not generally true when the momentum flux or energy flux is being considered. This is because the finite range of the intermolecular force field permits an interchange of momentum and energy between molecules that remain on opposite sides of the element. However, if the gas is dilute with $\delta \gg d$, these effects may be neglected.

The necessity for averages to be based on a sufficiently large sample, unless one resorts to time or ensemble averages, appears to impose a further restriction on expressions (1.17) and (1.18). The dimensions of the volume element in Fig. 1.6 should be large in comparison with the mean molecular spacing; i.e. $\bar{c}\,\Delta t$ should be much larger than δ. Also, the derivation of these expressions was based on the assumption that Δn is a constant. This is the case in an equilibrium gas but, in a non-equilibrium situation, the number of molecules scattered out of the class $\boldsymbol{c}$ by collisions may not equal the number scattered into the class. The analysis is therefore valid in a non-equilibrium gas only if Δt is much less than the mean collision time $1/\nu$. The gas must therefore be such that $\bar{c}\,\Delta t \gg \delta$.

Eqns (1.7) and (1.12) show that

$$\frac{\lambda}{\delta} = \frac{1}{\pi(\overline{c_r}/\overline{c'})}\left(\frac{\delta}{d}\right)^2. \tag{1.19}$$

The ratio $\overline{c_r}/\overline{c'}$ is of order unity and, consequently, $\lambda \gg \delta$ whenever the dilute gas condition ($\delta \gg d$) is satisfied. The ordering $\lambda \gg \delta \gg d$ is sometimes used as the definition of a dilute gas, but eqn (1.19) shows that eqn (1.9) is a sufficient condition.

Since $\boldsymbol{e}$ is a unit vector, the expression (1.17) can be written $(n\overline{Q\boldsymbol{c}}).\boldsymbol{e}$ and therefore defines the *flux vector* for the quantity Q as

$$n\overline{Q\boldsymbol{c}}. \tag{1.20}$$

The flux vector related to the transport of mass is obtained by setting Q equal to the molecular mass, to give,

$$nm\bar{\boldsymbol{c}} \quad \text{or} \quad \rho\bar{\boldsymbol{c}}.$$

The mean molecular velocity $\overline{\boldsymbol{c}}$ defines the macroscopic *stream* or *mean* or *mass velocity* which is denoted by $\boldsymbol{c}_0$, i.e.

$$\boldsymbol{c}_0 = \overline{\boldsymbol{c}} \,. \tag{1.21}$$

The velocity of a molecule relative to the stream velocity is called the *thermal* or *peculiar* or *random velocity* and is denoted by $\boldsymbol{c}'$, i.e.

$$\boldsymbol{c}' = \boldsymbol{c} - \boldsymbol{c}_0 \,. \tag{1.22}$$

Note that

$$\overline{\boldsymbol{c}'} = \overline{\boldsymbol{c}} - \overline{\boldsymbol{c}} = 0 \,,$$

so the mean thermal velocity is zero in a simple gas. The remaining macroscopic properties are defined by averages taken over the thermal velocities of the molecules. Therefore, in order to simplify the discussion of these quantities, we will now look at the element of gas from a frame of reference moving with the local stream velocity.

The flux vector for the thermal velocities is $n\overline{Q\,\boldsymbol{c}'}$ and an expression for the momentum transport by the peculiar or thermal motion is obtained by setting Q equal to $m\boldsymbol{c}'$. Since momentum is also a vector quantity, the resulting expression is a tensor with nine Cartesian components, called the *pressure tensor* $\boldsymbol{p}$, i.e.

$$\boldsymbol{p} = nm\overline{\boldsymbol{c}'\boldsymbol{c}'} = \rho\,\overline{\boldsymbol{c}'\boldsymbol{c}'} \tag{1.23}$$

and is best explained in terms of the separate components. Let u', v', and w' be the components of $\boldsymbol{c}'$ in the x-, y-, and z-directions and, as an example, consider the x-momentum flux in the y-direction. That is, set $Q = mu'$ and choose the element of area in the xz-plane so that its normal is in the y-direction. This component is

$$p_{xy} = \rho\,\overline{u'v'}$$

and the complete set is:

$$\begin{aligned}
&p_{xx} = \rho\overline{u'^2}, && p_{xy} = \rho\overline{u'v'}, && p_{xz} = \rho\overline{u'w'};\\
&p_{yx} = p_{xy}, && p_{yy} = \rho\overline{v'^2}, && p_{yz} = \rho\overline{v'w'};\\
&p_{zx} = p_{xz}, && p_{zy} = p_{yz}, \text{ and } && p_{zz} = \rho\overline{w'^2}.
\end{aligned} \tag{1.24}$$

A shorthand way of writing the nine equations (1.24) for the components of $\boldsymbol{p}$ is

$$p_{ij} = \rho\,\overline{c_i' c_j'} \,, \tag{1.24a}$$

where the subscripts i and j each range from 1 to 3. The values 1, 2, and 3 may be identified with the components along the x-, y-, and z-axes, respectively. That is,

$$c_1' \equiv u' \,, \quad c_2' \equiv v' \,, \quad c_3' \equiv w' \,.$$

The scalar *pressure* p is usually defined as the average of the three normal components of the pressure tensor; i.e.

$$p = \frac{1}{3}\rho\left(\overline{u'^2}+\overline{v'^2}+\overline{w'^2}\right) = \frac{1}{3}\rho\,\overline{c'^2}. \tag{1.25}$$

If the gas is in equilibrium, the three normal components are equal and the pressure is given by the product of the density and the mean value of the square of the thermal velocity components in any direction. Consider the case in which the gas is bounded by a solid surface and choose the reference direction normal to the surface. If the gas is also in equilibrium with the surface, the molecules reflected from the surface will be indistinguishable from the molecules that would come from an imaginary continuation of the gas on the other side of the surface. The scalar pressure p may then be identified with the normal force per unit area exerted by the gas on the surface. The latter quantity is usually of greater practical importance and, when there is an ambiguity, the term 'pressure' is applied to the normal force per unit area.

The *viscous stress tensor* τ is defined as the negative of the pressure tensor with the scalar pressure subtracted from the normal components. It is most conveniently represented in the component or subscript notation as

$$\tau \equiv \tau_{ij} = -\left(\rho\,\overline{c_i' c_j'} - \delta_{ij} p\right), \tag{1.26}$$

where δ_{ij} is the Kronecker delta such that

$$\delta_{ij} = 1 \quad \text{if} \quad i = j, \quad \text{and} \quad \delta_{ij} = 0 \quad \text{if} \quad i \neq j.$$

The average kinetic energy associated with the thermal or translational motion of a molecule is $\frac{1}{2}m\overline{c'^2}$, and the specific energy associated with this motion is

$$e_{tr} = \frac{1}{2}\overline{c'^2}. \tag{1.27}$$

This may be combined with eqn (1.25) to give

$$p = \frac{2}{3}\rho\, e_{tr},$$

which may be compared with the ideal gas equation of state

$$p = \rho R T = n k T. \tag{1.28}$$

Here k is the Boltzmann constant which is related to the universal gas constant $\mathcal{R}$ by $k = \mathcal{R}/\mathcal{N}$. Also $m = \mathcal{M}/\mathcal{N}$ and the ordinary gas constant $R = \mathcal{R}/\mathcal{M}$, so that $k = mR$. The *thermodynamic temperature* T is essentially an equilibrium gas property, but the above comparison shows that the ideal gas equation of state will apply to a dilute gas, even in a non-equilibrium situation, for a *translational kinetic temperature* T_{tr} defined by

$$\frac{3}{2}RT_{tr} = e_{tr} = \frac{1}{2}\overline{c'^2}$$

or

$$\frac{3}{2}kT_{tr} = \frac{1}{2}m\overline{c'^2} = \frac{1}{2}m\left(\overline{u'^2}+\overline{v'^2}+\overline{w'^2}\right). \tag{1.29}$$

If the temperature is calculated from the velocities of a set of molecules, it is preferable to have an expression in terms of averages over these velocities rather than over the thermal velocities. From eqn (1.22), $\overline{c'^2} = \overline{c^2} - 2\overline{\mathbf{c}}\mathbf{c}_0 + c_0^2$ and eqn (1.21) then enables eqn (1.29) to be written

$$\frac{3}{2}kT_{\mathrm{tr}} = \frac{1}{2}m\left(\overline{c^2} - c_0^2\right) = \frac{1}{2}m\left(\overline{u^2} + \overline{v^2} + \overline{w^2} - u_0^2 - v_0^2 - w_0^2\right). \qquad (1.29a)$$

The use of thermal velocities can also be avoided in the definition of the pressure tensor. For example,

$$p_{xy} = \rho\left(\overline{uv} - u_0 v_0\right).$$

Note also that separate translational kinetic temperatures may be defined for each component. For example,

$$kT_{\mathrm{tr}_x} = m\overline{u'^2} = p_{xx}/n \qquad (1.30)$$

and the departure of these component temperatures from T_{tr} provides a measure of the degree of translational non-equilibrium in a gas.

Monatomic molecules may generally be assumed to possess translational energy only. Therefore, in a monatomic gas, the translational temperature may be regarded simply as the *temperature*. However, diatomic and polyatomic molecules also possess internal energy associated with the rotational and vibrational energy modes. Since three degrees of freedom are associated with the translational mode, a temperature T_{int} for the internal modes may be defined consistently with the translational temperature (see eqn (1.29)) by

$$\frac{1}{2}\zeta RT_{\mathrm{int}} = e_{\mathrm{int}}. \qquad (1.31)$$

Here, ζ is the number of internal degrees of freedom and e_{int} is the specific energy associated with the internal modes. The principle of equipartition of energy means that the translational and internal temperatures must be equal in an equilibrium gas; the common value may then be identified with the thermodynamic temperature of the gas. An *overall kinetic temperature* T_{ov} may be defined for a non-equilibrium gas as the weighted mean of the translational and internal temperatures; i.e.

$$T_{\mathrm{ov}} = (3T_{\mathrm{tr}} + \zeta T_{\mathrm{int}})/(3 + \zeta). \qquad (1.32)$$

Note that the ideal gas equation of state does not apply to this temperature in a non-equilibrium situation.

Finally, the *heat flux vector* $\boldsymbol{q}$ is obtained by setting Q equal to the molecular energy $\frac{1}{2}mc'^2 + \varepsilon_{\mathrm{int}}$, i.e.

$$\boldsymbol{q} = \frac{1}{2}\rho\overline{c'^2\mathbf{c}'} + n\overline{\varepsilon_{\mathrm{int}}\mathbf{c}'}. \qquad (1.33)$$

$\varepsilon_{\mathrm{int}}$ is the internal energy of a single molecule and is related to e_{int} by

$$e_{\mathrm{int}} = \overline{\varepsilon_{\mathrm{int}}}/m.$$

The component of heat flux in the x-direction can be written

$$q_x = \frac{1}{2}\rho\,\overline{c'^2 u'} + n\,\overline{\varepsilon_{\text{int}} u'}\,. \tag{1.33a}$$

The use of thermal velocity components can again be avoided, to give

$$q_x = \frac{1}{2}\rho\left(\overline{c^2 u} - 2p_{xx}u_0 - 2p_{xy}v_0 - 2p_{xz}w_0 - \overline{c^2}u_0\right) + n\left(\overline{\varepsilon u} - \bar{\varepsilon}u_0\right). \tag{1.33b}$$

1.5 Extension to gas mixtures

Consider a gas mixture consisting of a total of s separate chemical species. Values pertaining to a particular species will be denoted by the subscripts p or q each of which can range from 1 to s. The overall number density n is obviously equal to the sum of the number densities of all the individual species; i.e.

$$n = \sum_{p=1}^{s} n_p\,. \tag{1.34}$$

Consider a collision between a molecule of species p and one of species q. The effective diameters are d_p and d_q, and the requirement for a collision is that the distance between their centres decreases to $(d_p + d_q)/2$. The total collision cross-section is therefore

$$\sigma_{Tpq} = \pi\,(d_p + d_q)^2/4 = \pi\,d_{pq}^2\,, \tag{1.35}$$

where

$$d_{pq} = (d_p + d_q)/2\,.$$

A molecule of species p may be taken as a test molecule and the molecules of species q as the field molecules in the analysis leading to eqn (1.10). The mean collision rate for a species p molecule with a species q molecule is, therefore,

$$\nu_{pq} = n_q\overline{\sigma_{Tpq} c_{rpq}}\,, \tag{1.36}$$

where c_{rpq} is the relative speed between the two molecules. If σ_{Tpq} is regarded as a constant, this becomes

$$\nu_{pq} = \pi\,d_{pq}^2\,n_q\,\overline{c_{rpq}}\,. \tag{1.36a}$$

The mean collision rate for species p molecules is obtained by summing over collision partners of all species; i.e.

$$\nu_p = \sum_{q=1}^{s}\left(n_q\,\overline{\sigma_{Tpq}\,c_{rpq}}\right). \tag{1.37}$$

A mean collision rate per molecule for the mixture may be defined by averaging over test particles of all species to give

$$\nu = \sum_{p=1}^{s}\left\{(n_p/n)\,\nu_p\right\}. \tag{1.38}$$

The number of collisions per unit time per unit volume between species p molecules and species q molecules is

$$n_p \nu_{pq}$$

or

$$n_p n_q \overline{\sigma_{T_{pq}} c_{r_{pq}}}. \tag{1.39}$$

Note that, when $p = q$, this expression counts each collision twice over. Eqn (1.39) may be summed over all species q molecules to give the total number of collisions per unit volume per unit time that involve species p molecules as $n_p \nu_p$. This, in turn may be summed over all type p molecules to determine the total number of collisions N_c per unit time per unit volume. Since all collisions are then counted twice over, the symmetry factor of one half is introduced to give

$$N_c = \frac{1}{2} \sum_{p=1}^{s} (n_p \nu_p) = \frac{1}{2} n \nu$$

as would be expected from eqn (1.11).

The mean free path for a species p molecule is equal to its mean thermal speed divided by its collision frequency, i.e.

$$\lambda_p = \left(\sum_{q=1}^{s} (n_q \overline{\sigma_{T_{pq}} c_{r_{pq}}} / \overline{c_p'}) \right)^{-1} \tag{1.40}$$

and the mean free path for the mixture is

$$\lambda = \overline{\lambda} = \sum_{p=1}^{s} \left\{ (n_p / n) \lambda_p \right\}. \tag{1.41}$$

The macroscopic density is equal to the sum of the individual species densities and can be written

$$\rho = \sum_{p=1}^{s} (m_p n_p) = n \overline{m}. \tag{1.42}$$

The stream velocity $\mathbf{c}_0$ has been defined for the simple gas such that $\rho \mathbf{c}_0$ is equal to the flux vector for the molecular mass. A similar procedure for the gas mixture yields

$$\boldsymbol{c}_0 = \frac{1}{\rho} \sum_{p=1}^{s} \left(m_p n_p \overline{\boldsymbol{c}_p} \right) = \overline{m \boldsymbol{c}} / \overline{m}. \tag{1.43}$$

For the mixture, $\mathbf{c}_0$ is called the *mass average velocity*. This is not the mean velocity, but a weighted mean with the weight of each molecule being proportional to its mass. The momentum of a gas is as if all the molecules move with $\mathbf{c}_0$ and it is this velocity that appears in the conservation equations.

The peculiar or thermal velocity $\mathbf{c}'$ of each molecule is again measured relative to $\mathbf{c}_0$, i.e.

$$\boldsymbol{c}' = \boldsymbol{c} - \boldsymbol{c}_0 ,$$

and the mean thermal velocity of species p is

$$\overline{\boldsymbol{c}_p'} = \overline{\boldsymbol{c}_p} - \boldsymbol{c}_0 . \tag{1.44}$$

Therefore, the mean thermal velocity of a particular species is equal to its mean velocity relative to the mass average velocity. This quantity is called the *diffusion velocity* and is denoted by $\boldsymbol{C}_p$. Therefore,

$$\boldsymbol{C}_p \equiv \overline{\boldsymbol{c}_p'} = \overline{\boldsymbol{c}_p} - \boldsymbol{c}_0 . \tag{1.45}$$

The definitions of the pressure tensor, the scalar pressure, the viscous stress tensor, the translational kinetic temperature, and the heat flux vector are rendered valid for the gas mixture simply by including the molecular mass within the averaging process. For example, the scalar pressure is

$$p = -\sum_{p=1}^{s} \frac{1}{3} m_p n_p \overline{c_p'^2} = -n \sum_{p=1}^{s} \frac{1}{3} (n_p/n) m_p \overline{c_p'^2} ,$$

or

$$p = \frac{1}{3} n m \overline{c'^2} . \tag{1.46}$$

The translational kinetic temperature is now defined by

$$\frac{3}{2} k T_{\mathrm{tr}} = \frac{1}{2} \overline{m c'^2} \tag{1.47}$$

and, in an equilibrium monatomic gas, T_{tr} can again be regarded simply as the temperature with the mixture obeying the ideal gas equation of state.

In a non-equilibrium situation, it is convenient to define separate 'species temperatures' to serve as a quantitative measure of the degree of non-equilibrium between the species. Eqn (1.47) can be written

$$\frac{3}{2} k T_{\mathrm{tr}} = \frac{1}{2} \sum_{p=1}^{s} \left\{ (n_p/n) m_p \overline{c_p'^2} \right\},$$

and the obvious definition of a species translational temperature is

$$\frac{3}{2} k T_{\mathrm{tr}p} = \frac{1}{2} m_p \overline{c_p'^2} . \tag{1.48}$$

However, the thermal velocity $\boldsymbol{c}_p'$ is measured relative to the mass average velocity $\boldsymbol{c}_0$ which involves all species. It is therefore desirable to relate the kinetic temperature defined by eqn (1.48) to one defined in a similar manner, but based on the *single species thermal velocity* $\mathbf{c}_p''$ measured relative to the average velocity $\overline{\boldsymbol{c}_p}$ of the species, i.e.

$$\boldsymbol{c}_p'' = \boldsymbol{c}_p - \overline{\boldsymbol{c}_p} . \tag{1.49}$$

This definition, together with eqn (1.45), enables eqn (1.48) to be written

$$\frac{3}{2} k T_{\mathrm{tr}p} = \frac{1}{2} m_p \overline{c_p''^2} + \frac{1}{2} m_p C_p^2 . \tag{1.50}$$

The kinetic temperature of species p, as defined by eqn (1.48), is therefore a measure of the sum of the 'single species thermal energy' and the 'kinetic energy of diffusion' of this species.

As in the simple gas case, rotational and vibrational kinetic temperatures may be defined for a gas containing diatomic or polyatomic molecules. Eqn (1.31) applies to the mixture as long as ζ is interpreted as the mean number of internal degrees of freedom. The single species translational temperature defined by eqn (1.48) may again be resolved into separate temperatures based on $\overline{u_p'^2}$, $\overline{v_p'^2}$, and $\overline{w_p'^2}$, but these must be assessed in conjunction with the component version of eqn (1.50) in order to determine the degree of translational non-equilibrium within each species. All the kinetic temperatures become equal and equivalent to the thermodynamic temperature when the gas is in equilibrium.

It is again possible to write the definitions of the second-order and third-order moments in terms of averages over the molecular velocities rather than the thermal velocities. The result for the temperature is

$$\tfrac{3}{2}kT_{\mathrm{tr}}/2 = \tfrac{1}{2}\left(\overline{mc^2} - \bar{m}c_0^2\right). \tag{1.51}$$

Similar results apply for all the components of the pressure and stress tensors, for example,

$$\tau_{xy} = -n(\overline{muv} - \bar{m}u_0v_0), \tag{1.52}$$

and the component of the heat flux vector in the x-direction is

$$q_x = n\left(\tfrac{1}{2}\overline{mc^2u} - p_{xx}u_0 - p_{xy}v_0 - p_{xz}w_0 - \tfrac{1}{2}\overline{mc^2}u_0 + \overline{\varepsilon u} - \bar{\varepsilon}u_0\right). \tag{1.53}$$

1.6 Molecular magnitudes

The accepted value of Avogadro's number for the number of molecules in one kmole[†] is 6.022137×10^{26}. The *standard number density* n_0 at a pressure of 1 atm (101,325 Pa) and a temperature of 0°C follows from eqn (1.28) as

$$n_0 = p/kT = 2.68666 \times 10^{25}\ \mathrm{m}^{-3}. \tag{1.54}$$

The standard number density can be regarded as a physical constant and the number 2.68684×10^{19} of molecules in a cubic centimetre is called *Loschmidt's number*. The volume of one kmole of an ideal gas under standard conditions is $\mathscr{N}/n_0$ or 22.414 m^3.

The discussion of the values of the other microscopic quantities will be based on air. For the purpose of this discussion, air will be assumed to be a simple gas of identical 'average air' molecules. The molecular weight of sea-level air is 28.96 kg kmole^{-1} and eqn (1.6) then gives 4.81×10^{-26} kg as the mass of a single molecule.

[†] The SI system of units is inconsistent in that the 'SI mole' is a 'gram mole' and a numerical factor must be applied to expressions that contain it when the basic units are employed throughout. This book therefore employs the 'kilogram mole' which is abbreviated to 'kmole'.

A difficulty arises in the definition of the molecular diameter and mean free path in that the intermolecular force at moderate to high separation distances declines as some inverse power of this distance. An arbitrary collisional cut-off in either distance or deflection angle is necessary for realistic molecular models, and it is not then possible to unambiguously define the diameter and mean free path. The conventional solution to this problem is to deduce a molecular diameter from a comparison of the Chapman–Enskog result for the viscosity coefficient in a hard sphere gas with the measured coefficient of viscosity in the real gas. However, the hard sphere gas has a fixed cross-section and it will be shown in subsequent chapters that the accurate simulation of real gas flows requires a molecular model with a cross-section that varies with the relative velocity between the molecules. The cross-section change at the molecular level is related to the variation of the coefficient of viscosity with temperature at the continuum level. None of the molecular models of classical kinetic theory has both a bounded cross-section and the capability to reproduce the viscosity–temperature variation of a real gas. This difficulty has been overcome through the introduction (Bird, 1981) of the variable hard sphere or VHS model. This model incorporates the hard sphere scattering law for collisions, but its cross-section is a function of the relative translational energy in the collision. It will be shown in Chapter 4 that a viscosity coefficient proportional to temperature to the power ω is obtained if the average cross-section is proportional to temperature to the power $-(\omega-\frac{1}{2})$.

An average VHS molecular diameter may be obtained from a comparison of the measured coefficient of viscosity with the theoretical result (Bird 1981). For air, this gives $\bar{d} = 4.15\times10^{-10}$ m for the average diameter at 273 K and this is inversely proportional to the one-eighth power of temperature. The mean molecular spacing under standard conditions is obtained from n_0 through eqn (1.7) as $\delta_0 = 3.3\times10^{-9}$ m. Air under standard conditions therefore barely satisfies the dilute gas condition that $\delta \gg d$. The third distance of interest is the mean free path defined by eqn (1.12). It will be found that the mean free path in an equilibrium VHS gas is

$$\lambda = 1/(\sqrt{2}\,\pi\bar{d}^{\,2}n). \tag{1.55}$$

The viscosity coefficient is very nearly proportional to temperature raised to the power ¾. The mean free path in equilibrium air under standard conditions is therefore 4.9×10^{-8} m and this is inversely proportional to number density and proportional to the ¼ power of the temperature. The mean free path is independent of temperature for the special case of a hard sphere gas which corresponds to $\omega = \frac{1}{2}$ and has a constant diameter.

The mean square molecular speed is given by eqns (1.28) and (1.29) as

$$\overline{c'^2} = 3p/\rho = 3RT = 3kT/m. \tag{1.56}$$

The root mean square molecular speed $c_s' = \left(\overline{c'^2}\right)^{1/2}$ therefore differs from the speed of sound in the gas (which is given by $a^2 = \gamma RT$) only by a constant of order unity. Using the previously quoted value for the average mass of an air molecule, we obtain 485 m s^{-1} for the root mean square speed of the molecules in air at 0°C. For an equilibrium gas, we will find that the mean thermal speed $\overline{c}$ is $(8/[3\pi])^{1/2}$ times c_s', and the mean magnitude of the relative velocity of colliding molecules is $\sqrt{2}$ times the mean thermal speed. The mean collision rate of a molecule in equilibrium air under standard conditions is then given by eqn (1.10) as 7.3×10^9 s^{-1}, and the total collision rate per unit volume follows from eqn (1.11) as 9.8×10^{34} m^{-3}s^{-1}.

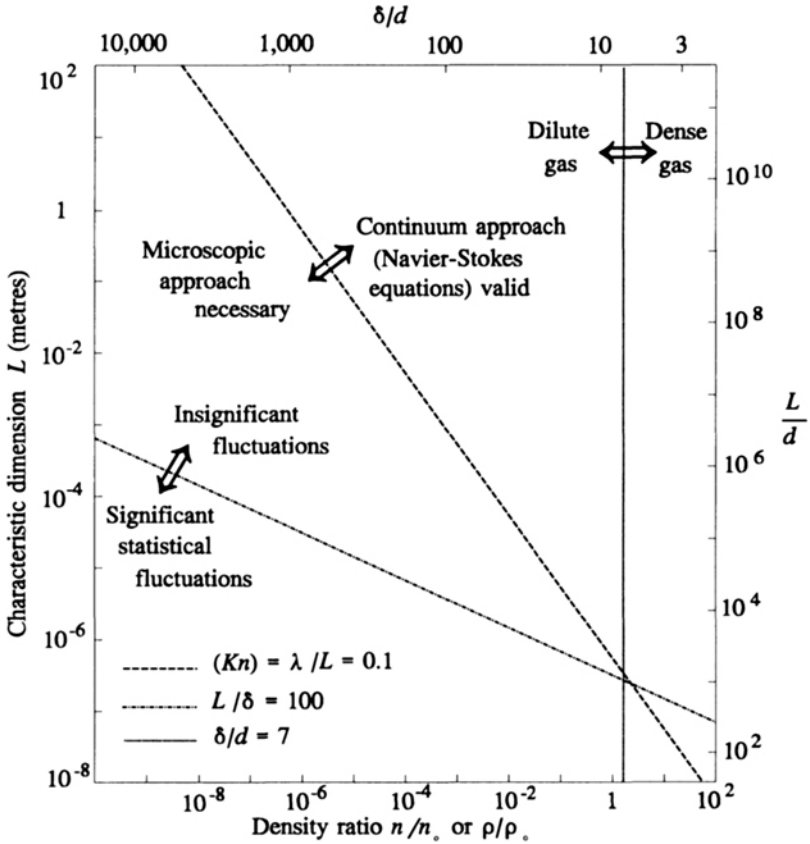


Fig. 1.7 Effective limits of major approximations.

We are now in a position to define the limits of validity of the dilute gas approximation, the continuum approach, and the neglect of statistical fluctuations. For the purposes of this study, the molecular diameter will be fixed at 4×10^{-10} m. The limits are conveniently expressed as functions of the gas density ρ and the characteristic dimension of the flow L. The density may be normalized by the density ρ_0 under standard conditions, but L is best retained as a dimensioned quantity. In the context of a double logarithmic plot of L versus ρ/ρ_0, the three limits may be defined by straight lines, as shown in Fig. 1.7.

The dilute gas assumption requires that $\delta/d >> 1$, and $\delta/d = 7$ has been chosen as the limit. Since both δ and d are independent of L, the line is vertical and a scale for δ/d has been set along the upper edge of Fig. 1.7.

The validity of the continuum approach has been identified with the validity of the Navier–Stokes equations. This requires that the Knudsen number $(Kn) = \lambda/L$ should be small compared with unity, and $(Kn) = 0.1$ has been chosen as the limit. This is a reasonable choice as long as L is chosen such that the Knudsen number is the local Knudsen number, as discussed in §1.2.

The macroscopic flow properties are subject to statistical fluctuations, as discussed in §1.4. The criterion for the onset of significant fluctuations has been chosen as $L/\delta = 100$. This corresponds to a value of L that is ten times as large as the side of a cubic element that contains 1,000 molecules. Figure 1.5 shows that the density in such an element would be subject to statistical fluctuations with a standard deviation of approximately three per cent.

A feature of Fig. 1.7 is that the lines describing the three limits very nearly intersect at a single point. This result would not be substantially altered by any reasonable changes in the typical dimensions and the criteria that define the limits. One consequence is that the continuum breakdown limit always lies between the dilute gas limit and the limit for significant fluctuations. Therefore as the density and/or the characteristic dimension are reduced in a dilute gas, the Navier–Stokes equations lose their validity before the level of statistical fluctuations becomes significant. The more rarefied the gas, the wider the margin between these limits. On the other hand, a significant level of fluctuation may be present in a dense gas even when the Navier–Stokes equations are valid. For example, the theory of Brownian motion is partially based on these equations although the phenomenon is, itself, a manifestation of significant statistical fluctuations.

The fact that fluctuations are more significant in dense gases indicates that, although the molecules become more closely packed as the density increases, the number of molecules that are involved in typical flows actually decreases as the density increases. There are several other functions of the density ratio that clarify this point.

Again using the hard sphere diameter of 4×10^{-10} m, the fraction of space that is occupied by molecules is

$$\pi d^3 n/6 = 0.0009(n/n_0). \tag{1.57}$$

However, as the density changes, the flow gradients tend to scale with the mean free path and the number of molecules within one cubic mean free path is

$$n\lambda^3 = 3856(n/n_0)^{-2}. \tag{1.58}$$

This means that, for a given flow, the total number of molecules within the flowfield increases as the density decreases. The results from eqns (1.57) and (1.58), together with the ratio of molecular diameter to mean free path, are plotted against the density ratio in Fig. 1.8. This figure includes an upper scale that relates the density ratio to the altitude in the standard atmosphere.

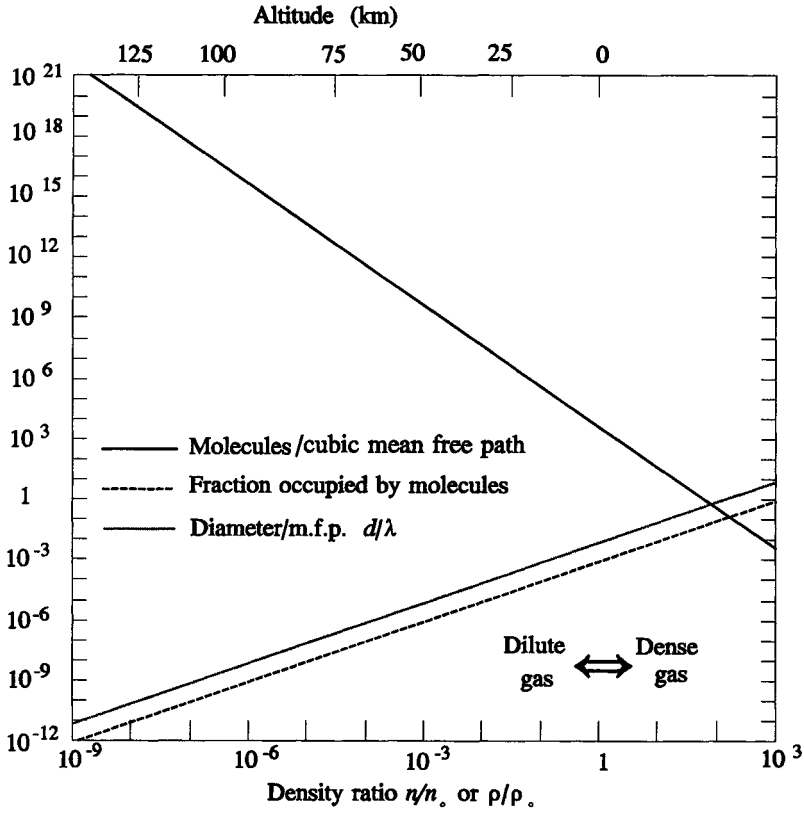


Fig. 1.8 Some consequences of molecular size.

1.7 Real gas effects

The discussion in the preceding sections has implicitly assumed that the molecules are adequately described by classical or Newtonian physics. Moreover, the possibility that molecules change their identities as a result of collisions has not been taken into account. Real gas models must include chemical reactions, and also the absorption and emission of radiation.

Quantum mechanics provides a more accurate description and shows that there is a limit on the precision with which the molecular positions and momenta may be specified. The Heisenburg uncertainty principle states that the product of the uncertainty $|\Delta r|$ in position and $|\Delta mc|$ in momentum is of the order of Planck's constant, i.e.

$$|\Delta r|\,|\Delta mc| \sim h. \tag{1.59}$$

For the classical model to be adequate, the uncertainties in position and momentum should be much smaller than the mean molecular spacing and the mean magnitude of the molecular momentum, respectively. This requires

$$\delta m\bar{c} \gg h \tag{1.60}$$

and means that δ should be much larger than the mean de Broglie wavelength $(h/m\bar{c})$ of the molecules. The condition for gross diffraction effects to be absent from collisions is that the molecular diameter d should be large compared with the de Broglie wavelength. The second condition is the more restrictive in a dilute gas, but, at this stage, our primary concern is with the general molecular description rather than the detail of the collision processes.

A measure of molecular speed may be obtained from eqn (1.56) and, using eqn (1.7), eqn (1.60) becomes

$$(3mkT)^{1/2}/(n^{1/3}h) \gg 1. \tag{1.61}$$

The substitution of the 'average air' mass at standard temperature and density into the expression on the left-hand side of this equation leads to a value of 117.4. Even at low temperatures, the dilute gas assumption would break down well before the density increased sufficiently to violate eqn (1.61). Representative data on gases other than air is presented in Appendix A. The only circumstance in which a quantum description becomes necessary for the general molecular model is when a light gas such as hydrogen or helium is at an extremely low temperature.

A further condition for the validity of the classical model is that the number of available quantum states should be very large in comparison with the number of molecules. The standard result for the ratio of the number of available translational states to the number of molecules is, to within a numerical factor of order unity, the cube of the expression on the left-hand side of eqn (1.61). The classical approximation is therefore

applicable to the translational motion which contributes three degrees of freedom to the gas. As discussed in §1.4, each fully excited degree of freedom contains a specific energy of $\frac{1}{2}kT$ in an equilibrium gas.

Quantum considerations must be taken into account when discussing the internal degrees of freedom associated with rotation and vibration of the molecules. The rotational mode has an angular momentum $I\omega$ and rotational energy $\frac{1}{2}I\omega^2$, where I is the moment of inertia about the axis of rotation and ω is the angular velocity. Rotation can be neglected for monatomic molecules and for diatomic molecules about the internuclear axis. This is because the moment of inertia is so small that the minimum quantum of angular momentum would involve rotational energy far higher than that associated with the temperatures of interest. Diatomic molecules have two degrees of freedom associated with the axes normal to the internuclear axis. The spacing of the rotational states is of the order of $k\Theta_{rot}$, where Θ_{rot} is the characteristic temperature of rotation. The rotational mode is not fully excited until the temperature is large compared with Θ_{rot}. Appendix A shows that the characteristic rotational temperatures of air molecules are between two and three degrees Kelvin, so that rotation is fully excited in air at normal temperatures. Triatomic and polyatomic molecules generally have three degrees of freedom of rotation.

Two degrees of freedom may be associated with the vibrational motion of diatomic molecules. However, the vibrational levels are widely spaced and the energy that would be assigned to a classical molecule with two degrees of freedom is truncated to that of the highest available level. The characteristic temperatures of vibration Θ_v for air molecules are between two and three thousand degrees Kelvin. Therefore, at normal temperatures, most air molecules are in the ground state, very few have vibrational energy equal to $k\theta_v$, and there are almost none in the higher levels. This means that vibration may be neglected at temperatures less than about 1000 K. A large number of vibrational modes are present in polyatomic gases, and some modes in gases such as carbon dioxide have characteristic temperatures that are comparable with the normal temperature.

The specific heat ratio γ is related to the number of excited degrees of freedom ξ by

$$\gamma = c_p/c_V = (\xi + 2)/\xi, \tag{1.62}$$

and γ must be treated as a variable when T is of the order of Θ_v. This applies to air at temperatures above 1,000 K and γ drops below the usual value of 7/5. If vibration was fully excited in air γ would be 9/7 but, before this is attained, a significant fraction of molecular oxygen and nitrogen is dissociated into atoms. The activation energy for the dissociation of oxygen is 5.12 ev and that for nitrogen is 9.76 ev. A characteristic temperature for dissociation Θ_d is defined by dividing the activation energy by the Boltzmann constant k. This gives 59,500 K for oxygen and 113,000 K for nitrogen. These large values are misleading in that there is significant dissociation at temperatures that are small compared with Θ_d. This is due, in part, to there being a significant number of collisions that involve

relative energies very much higher than kT. However, the main reason is that the reverse or recombination reaction cannot occur in a binary collision. It is an exothermic reaction and it is not possible to simultaneously satisfy the momentum and energy constraints if there is just one post-collision molecule. A third particle must participate in the collision, and three-body or ternary collisions are very rare events in a dilute gas. The dissociation–recombination reaction set is further complicated by the formation of a small amount of nitric oxide. This is important for the ionization reactions that become significant at temperatures of the order of 10,000 K. The ionization potential of NO is 9.34 ev and this is much less than the 12.3 ev for O_2, 15.7 ev for N_2, 13.7 ev for O, and 14.6 ev for N.

The electrons are negatively charged, are orders of magnitude lighter than the other particles, and interact with the positively charged ions through long-range Coulomb interactions. A number of the key approximations that are made for neutral gases cease to be valid when ionisation is present and additional modes of propagation of disturbances become possible. When there is a significant degree of ionization, the field of study changes from kinetic theory and gas dynamics to plasma physics. To the extent that this book deals at all with charged gases, it will be concerned only with very small degrees of ionization that allow simplified models of the plasma effects.

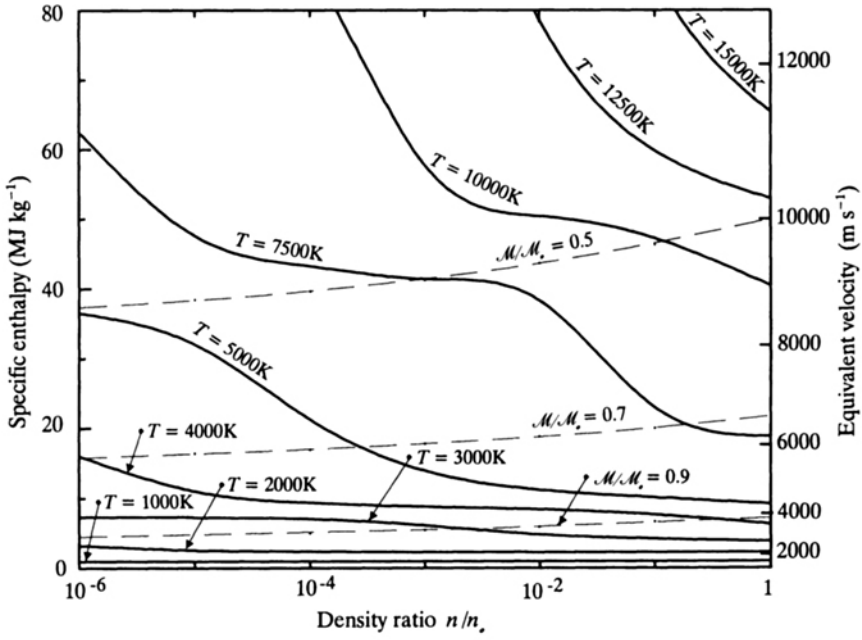


Fig. 1.9 Temperature and molecular weight as a function of enthalpy in equilibrium air.

The large amount of energy required by the dissociation reactions is made clear in Fig. 1.9 which shows the enthalpy of the gas that is required to reach various values of temperature in an equilibrium gas. The specific enthalpy $h = c_p T$ and, at temperatures of the order of the standard temperature, c_p is nearly constant with a value of 1002 Jkg^{-1}. The numerical values of the specific enthalpy are unchanged when expressed in units of m^2s^{-2}, and an equivalent velocity scale has been included in the figure. The energy of dissociation is primarily responsible for the order of magnitude increase in c_p in real air. The higher enthalpy for a given temperature at the lower densities is due to the ratio of ternary to binary collisions being proportional to density, and the consequent reduced rate of recombinations at low densities leads to a higher degree of equilibrium dissociation. A large number of reactions occur in high-temperature air, and this is responsible for the irregularity of the contours of constant temperature in Fig. 1.9. The number of particles increases in both dissociation and ionization reactions. This means that the molecular weight also changes, and some contours of constant molecular weight are shown in Fig. 1.9.

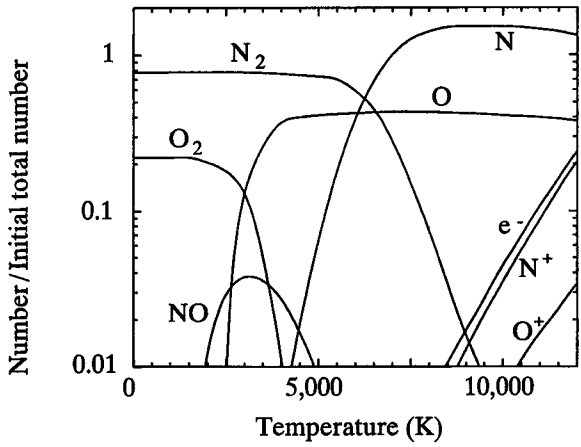


Fig. 1.10 Equilibrium composition of air at $n/n_0 = 10^{-2}$.

The equilibrium composition of air as a function of temperature is shown in Fig. 1.10 for a density ratio $n/n_0 = 10^{-2}$. Statistical mechanics enables these equilibrium properties to be readily calculated from the partition functions. However, when an equilibrium gas is subjected to a flow disturbance, a new equilibrium state can only be attained through intermolecular collisions and these may not be sufficiently rapid for each gas element to remain in equilibrium as it passes through the flow. We are

then faced with a non-equilibrium flow in which the finite rates of the chemical reactions are coupled with the flow changes. The concept of local thermodynamic equilibrium was introduced in §1.3 in the context of the translational modes. Chemical non-equilibrium and the non-equilibrium between the translational and internal modes almost always accompany translational non-equilibrium, and frequently occur even when the translational modes are very close to equilibrium. The thermodynamic definition of temperature has no meaning in a non-equilibrium flow and, apart from the limiting case in which the internal modes and chemistry are *frozen*, such flows are non-isentropic. The analysis of non-equilibrium flows must be based on a knowledge of the details of the physical processes.

The time that is required for the gas to come to equilibrium is called the *relaxation time*. If there is a finite stream velocity, an element of gas moves a finite distance during this time and a relaxation distance is established. Relaxation times are generally proportional to the mean collision time and relaxation distances are generally proportional to the mean free path. The non-equilibrium regions of typical gas flows become more extensive as the Knudsen number increases and, in highly rarefied gases, the whole flowfield may be very far from equilibrium. The kinetic temperatures based on the various energy modes are then very different.

The translational modes are generally assumed to adjust to the new conditions within a few mean collision times, although it should be noted that relaxation times are based on the initial gradient of the change and a much longer time is needed for a full accommodation to the new conditions. In a region of translational non-equilibrium, the kinetic temperatures based on the separate components of the molecular velocities, as defined in eqn (1.30), will generally differ. For example, the kinetic temperature based on the velocity components in the direction of the flow within a strong normal shock wave may be double the kinetic temperature based on the velocity components normal to the flow direction. The relaxation times for the rotational modes are generally of the order of five to ten mean collision times, so that the adjustment is not much slower than that for the translation. On the other hand, the relaxation times for vibration are, at normal temperatures, at least two or three orders of magnitude slower than those for translation and rotation, and vibrational non-equilibrium is the rule rather than the exception.

Dissociation and recombination are chemical reactions and data for the speed of particular reactions is available in the form of rate coefficients at the continuum level and cross-sections for the reactions at the particle level. Alternatively, dissociation can be regarded as vibrational excitation to the level at which the atoms split apart, and the dissociation rate is set by the vibrational relaxation rate. Vibrational relaxation is much faster at very high temperatures and, together with the associated dissociation, it can then proceed more rapidly than the rotational relaxation. As noted earlier, recombination is an extremely slow process and the dissociation level in an expanding flow may often be regarded as frozen.

Thermal radiation is an additional mechanism for the transfer of energy in high-temperature gases and can occur even in collisionless flows. For equilibrium radiation, the energy absorbed at each frequency is equal to that radiated but, like the reactions, the radiation can be non-equilibrium. The most important mechanism for the emission of radiation is generally the *bound–bound* transitions between the quantum energy levels of the atoms and molecules. Since the energy associated with each level is fixed within narrow bounds, the energy in these transitions also falls within a narrow energy range and the radiation gives rise to distinct spectral lines. Radiation from rotational transitions has the lowest energy and is usually within the infrared band. This radiation is emitted only by molecules with dissimilar atoms and is almost absent from high-temperature air. Vibrational transitions are responsible for most of the radiation from high-temperature air and many of the lines fall within the visible region of the spectrum. The absorption of radiation can result in dissociation or recombination and this, together with emission from the reverse reactions, is called *bound–free* radiation. *Free–free* radiation is produced by changes in the energy of free electrons and is generally associated with problems in plasma physics that are outside the scope of this book.

Radiation theory may be based on a continuum or a particle approach. The continuum theory employs emission and absorption coefficients, and is concerned with the optical thickness of the gas. The particle approach to line radiation is based on the probabilities of the individual radiation events, and the Einstein coefficients are the fundamental quantities.

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